Macro-level actions for efficient & reliable Web UI Agents

Повышение эффективности веб-агентов через высокоуровневые действия

PhD research proposal
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Web UI automation in 2026

What? Allow AI agents to interact with legacy web systems autonomously or in HITL setups.
Adoption: rapid, via AI chats (Claude Cowork) or agentic browsers (Comet)
SOTA: multimodal perception over foundation models (LLM / VLM) [1]
Research goal: increase efficiency of LLM / VLM-driven approaches with RPA-like actions

Approach	Setup	Cost	stability	generalizability
Scripted / RPA	$$	$	-	-
Imitation learninig	$$	$$	+	±
RL agents	$$$	$$	+	-
LLM (DOM / AXTree)	$$	$$$	+	+
VLM Agents	$$	$$$	++	++

Open challenges

The current web UI agents' low-level point-and-click / type actions and often fail in real-world applications:

1. Long-horizon planning / compounding errors => silent failures
2. Context bloat, high compute cost & latency
3. Perceptual grounding

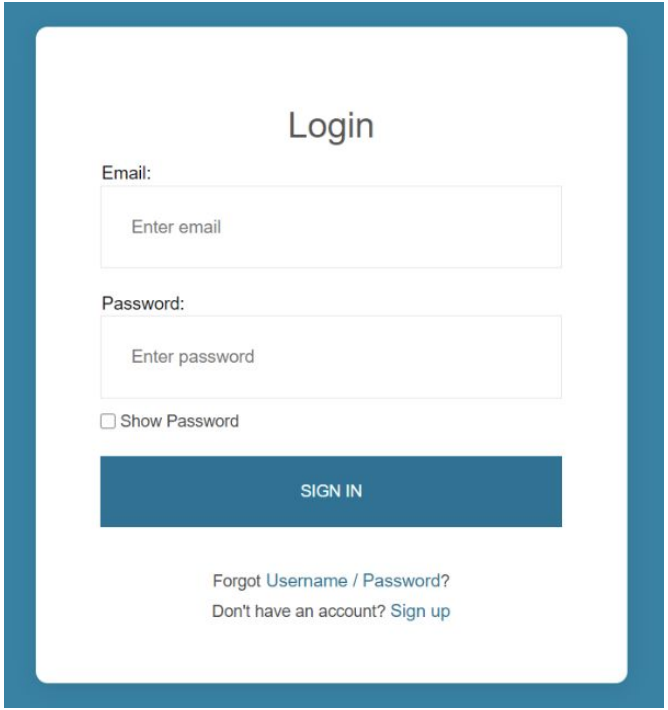
Extra challenges (not directly targeted):

4. Security (prompt injections, runaway autonomy)
5. CAPTCHA / anti-bot

Proposed concept

Hybrid LM / RPA automation via “macro-level actions”: instead of deciding where to click at each step, launch deterministic flow (e.g. *login*) with explicit pre- and post-condition checks. We expect to:

- Reduce step count (and LLM calls)
- Increases success rate in multi-step tasks
- Decrease silent failures via deterministic checks
- Lower token usage



Research questions & methodology

Question	Methodology
(prerequisite) What front-end patterns cause agent failures?	Augmented extension of Online-Mind2Web / Odysseus benchmarks using code-level mutations
Q1. Do pre-defined macro actions increase multi-step success rate?	Benchmark OSS agents with and without manually scripted tools
Q2. Can macro actions be automatically generated on the fly?	Supervised Fine-Tuning (SFT) / Reinforcement Learning (RLVR)
Q3. Can relevant macro actions be retrieved when needed, in a generalizable way?	RAG / Graph RAG
Q4. Can macro actions be automatically generated from source code at build time?	Benchmark pass@k for generating pre-defined action flows in OSS front-end projects

References

[1] P. J. Sager et al., "A Comprehensive Survey of Agents for Computer Use: Foundations, Challenges, and Future Directions," *Journal of Artificial Intelligence Research*, vol. 85, Mar. 2026, doi: 10.1613/jair.1.19490.

[2] T. Xue et al., "An Illusion of Progress? Assessing the Current State of Web Agents," 2025, *arXiv:2504.01382*.

[3] K. Yu, N. Yu, H. Wang, R. Yang, and H. Zhang, "How do Visual Attributes Influence Web Agents? A Comprehensive Evaluation of User Interface Design Factors," 2026, *arXiv:2601.21961*.